

Experiment 6

1. Experiment Title

Use String Functions and Regular Expressions (Regex) for Email Validation, Data Extraction, and Text Analysis

2. Software/Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- CSS3
- JavaScript (ES6)
- Live Server Extension

3. Theory

3.1 Strings in JavaScript

A string is a sequence of characters used to represent text in JavaScript. JavaScript provides several built-in methods for manipulating and analyzing strings.

Commonly used string functions include:

- `trim()` – Removes whitespace from the beginning and end of a string.
- `toUpperCase()` – Converts a string into uppercase letters.
- `toLowerCase()` – Converts a string into lowercase letters.
- `split()` – Divides a string into an array of substrings.
- `match()` – Searches a string using a regular expression.
- `length` – Returns the number of characters in a string.

3.2 Regular Expressions

A **Regular Expression (Regex)** is a sequence of characters that defines a search pattern. Regular expressions are useful for searching, matching, validating, and extracting specific information from text.

JavaScript provides the **RegExp** object and methods such as **test()** and **match()** for working with regular expressions.

3.3 Email Validation

Regular expressions can be used to validate whether an entered email address follows a valid general format.

The pattern used in this experiment is:

```
/^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/
```

It checks for:

- Username
- @ symbol
- Domain name
- Domain extension

3.4 Data Extraction Using Regex

Regular expressions can also be used to extract useful information from a larger text.

In this experiment, the program extracts:

- Email addresses
- 10-digit phone numbers
- Words

3.5 Text Analysis

String functions and regular expressions can be combined to analyze text.

The program determines:

- Number of characters
- Number of words
- Number of sentences
- Number of vowels
- Number of digits
- Uppercase version of the text
- Lowercase version of the text

5. Output

Email Validation

chaudhariradhika1150@gmail.com

Validate Email

✓ Valid Email Address

Email: chaudhariradhika1150@gmail.com

Text Analysis

JavaScript is a powerful programming language. JavaScript is widely used for web development.

Analyze Text

Characters: 93

Words: 13

Sentences: 2

Vowels: 29

Digits: 0

Uppercase Text:
JAVASCRIPT IS A POWERFUL PROGRAMMING LANGUAGE. JAVASCRIPT IS WIDELY USED FOR WEB DEVELOPMENT.

Lowercase Text:
javascript is a powerful programming language. javascript is widely used for web development.

Data Extraction

Contact us at support@example.com or admin@gmail.com.
Call us at 9876543210 or 9123456789.

Extract EmailsExtract Phone NumbersExtract Words

Total Words: 14

Extracted Words:

ContactusatsupportexamplecomoradmingmailcomCallusat or

Email Validation

Radhika1150@gmail

Validate Email

✗ Invalid Email Address

Please enter a valid email address.

6. Result/Conclusion

The experiment was successfully implemented using **HTML5, CSS3, and JavaScript**. String functions such as `trim()`, `split()`, `match()`, `toUpperCase()`, and `toLowerCase()` were used for text manipulation and analysis. Regular expressions were used for email validation, data extraction, and identification of specific patterns such as phone numbers, words, vowels, and digits. The experiment provided practical understanding of string processing and Regex in JavaScript and demonstrated their application in interactive web development.